Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 28)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

0 1 . 2 Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

0 1 . 3 Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

0 1 . 4 Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

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0 2 The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 2 . 1 Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer ____ thousand tonnes

0 2 . 2 Calculate the maximum catch that would have kept the stock constant.

[1 mark]

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Answer [ ] thousand tonnes

0 2 . 3 Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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0 2 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort .....

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Mean fish size and mean age .....

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0 3 A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

0 3 . 1 Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

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0 3 . 2 Explain the changes shown in Table 1.

[5 marks]

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0 3 . 3 Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

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0 3 . 4 Describe the early stages of a lithosere.

[1 mark]

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0 4 . 1 Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest ☐

B Oceanic islands ☐

C Temperate broadleaf woodland ☐

D Mangroves ☐

0 4 . 2 Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

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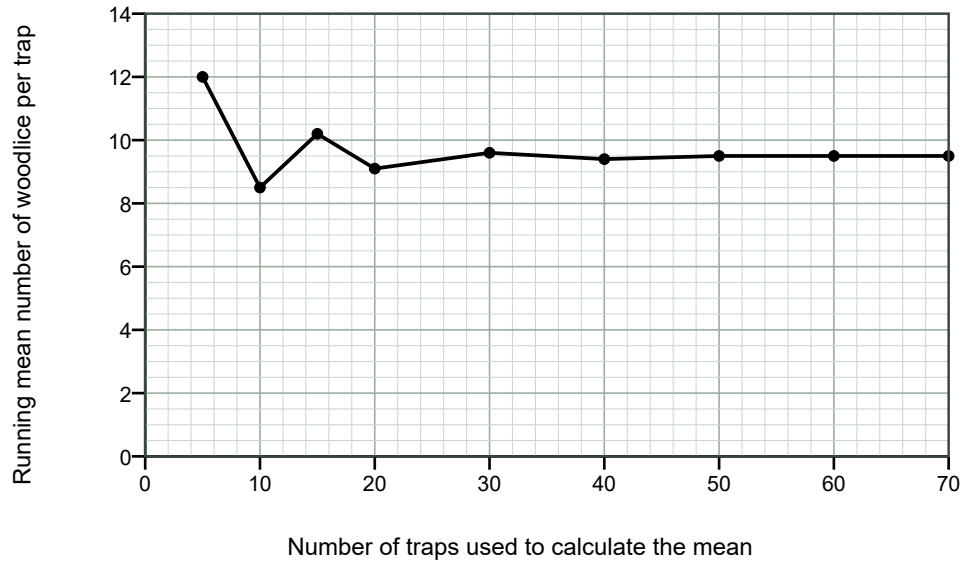
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0 5 Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 5 . 1 Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

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0 5 . 2 Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

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0 5 . 3 The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

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0 5 . 4 Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

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0 6 . 1 Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

0 6 . 2 Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

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0 6 . 3 Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

0 6 . 4 Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

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0 7 Table 1 shows the energy inputs and food energy outputs of three farming systems, in GJ ha^{-1} yr^{-1}.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Subsistence maize, hand cultivated	0.6	9.0
Intensive wheat, UK	18.0	54.0
Intensive beef, feedlot	40.0	12.0

0 7 . 1 Calculate the energy ratio (output / input) for each system in Table 1.

[3 marks]

Answer

0 7 . 2 Explain what is meant by an **energy subsidy**, and use your answers to explain why the intensive systems have much lower energy ratios.

[3 marks]

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0 7 . 3 Explain **three** ways a farmer can reduce the energy losses in a food production chain.

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0 7 . 4 Explain **two** advantages of genetic modification over conventional selective breeding.

[2 marks]

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0 7 . 5 Explain **four** concerns about the use of GM crops.

[4 marks]

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0 8 . 1 Describe how agriculture alters the hydrological cycle.

[3 marks]

0 8 . 2 Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

Organic food

0 8 . 3 Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

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0 9 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 9 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 9 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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1 0 . 1 Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

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1 0 . 2 State what is meant by **biodiversity**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

OR

1 1 . 2 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS